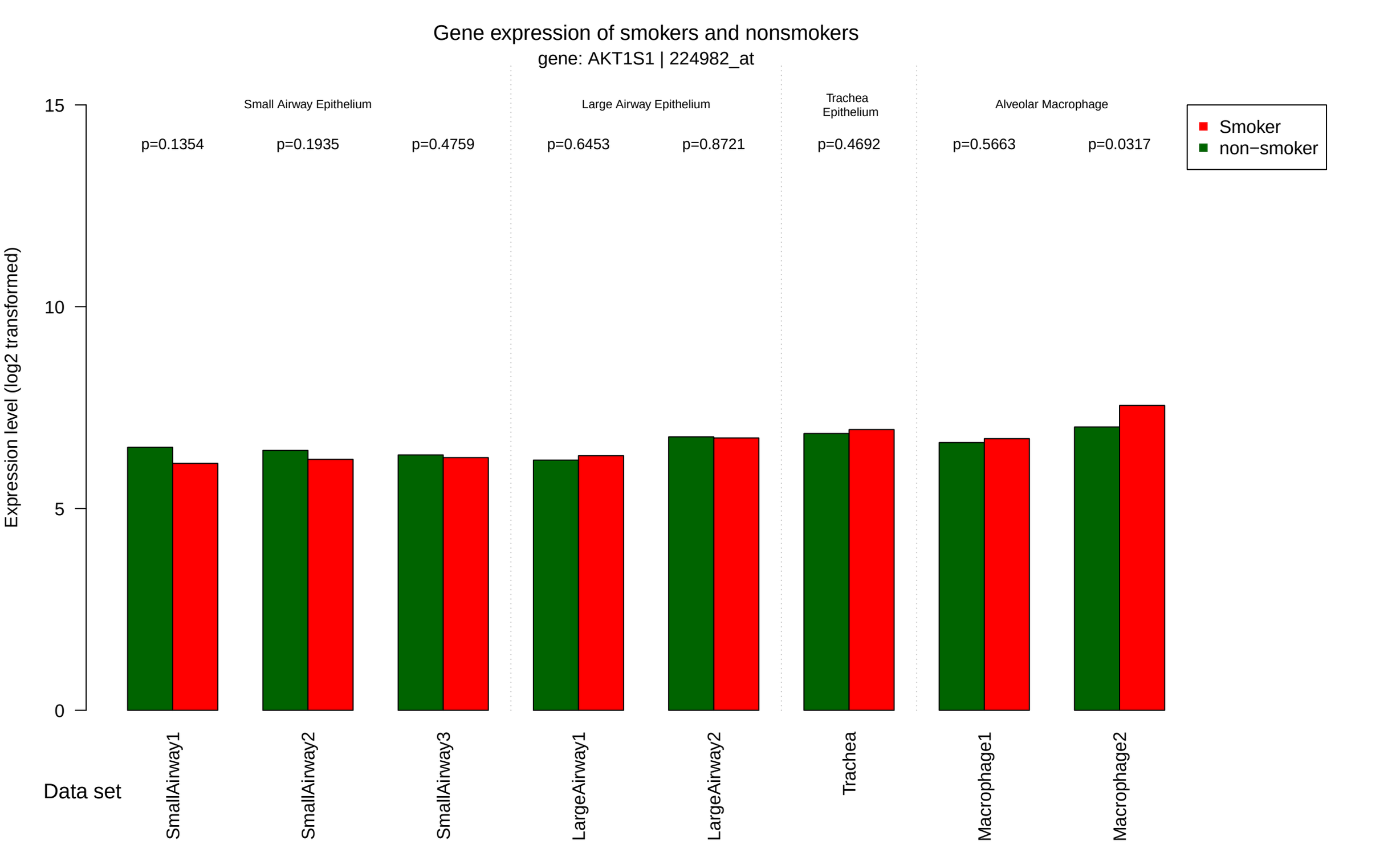
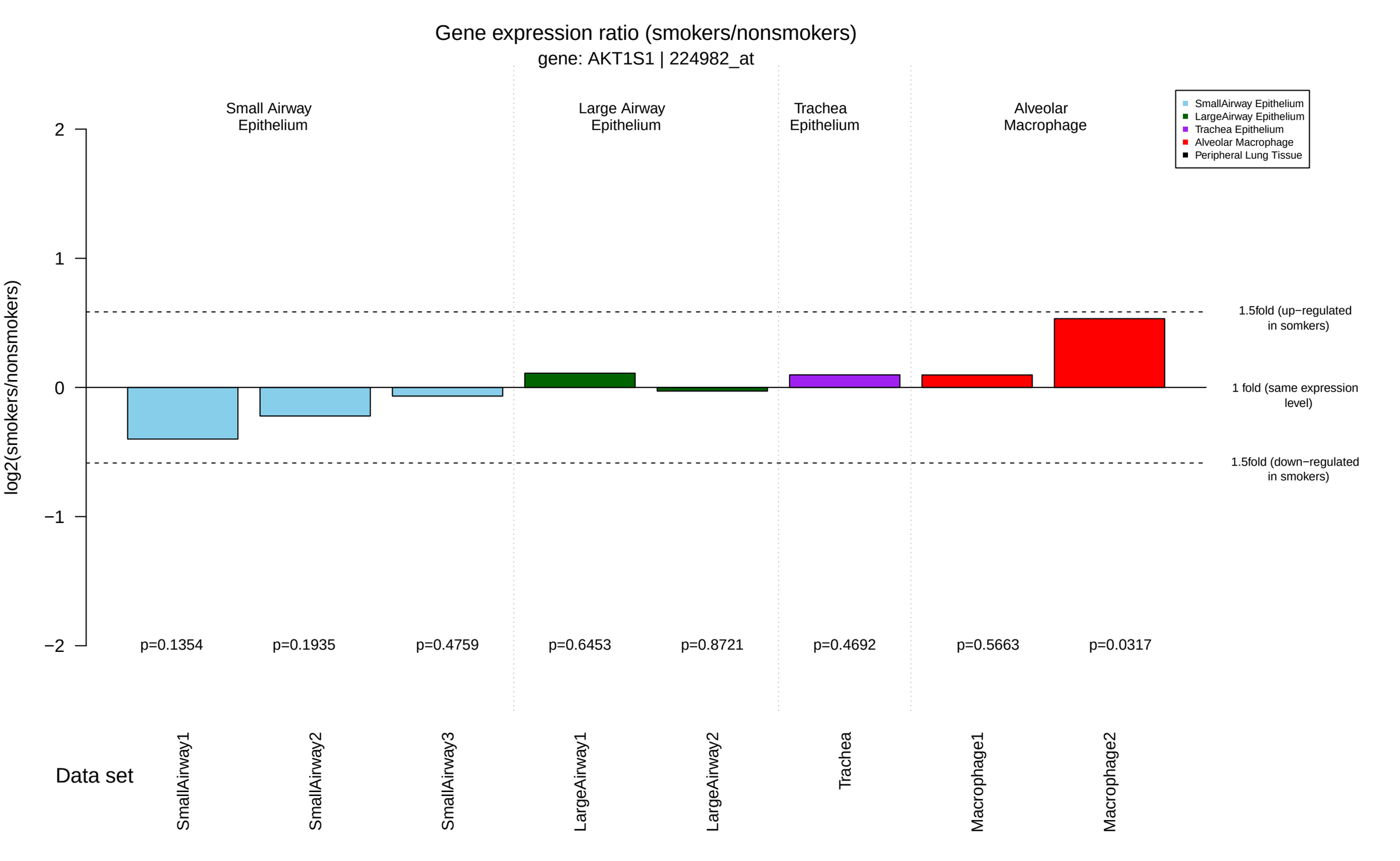
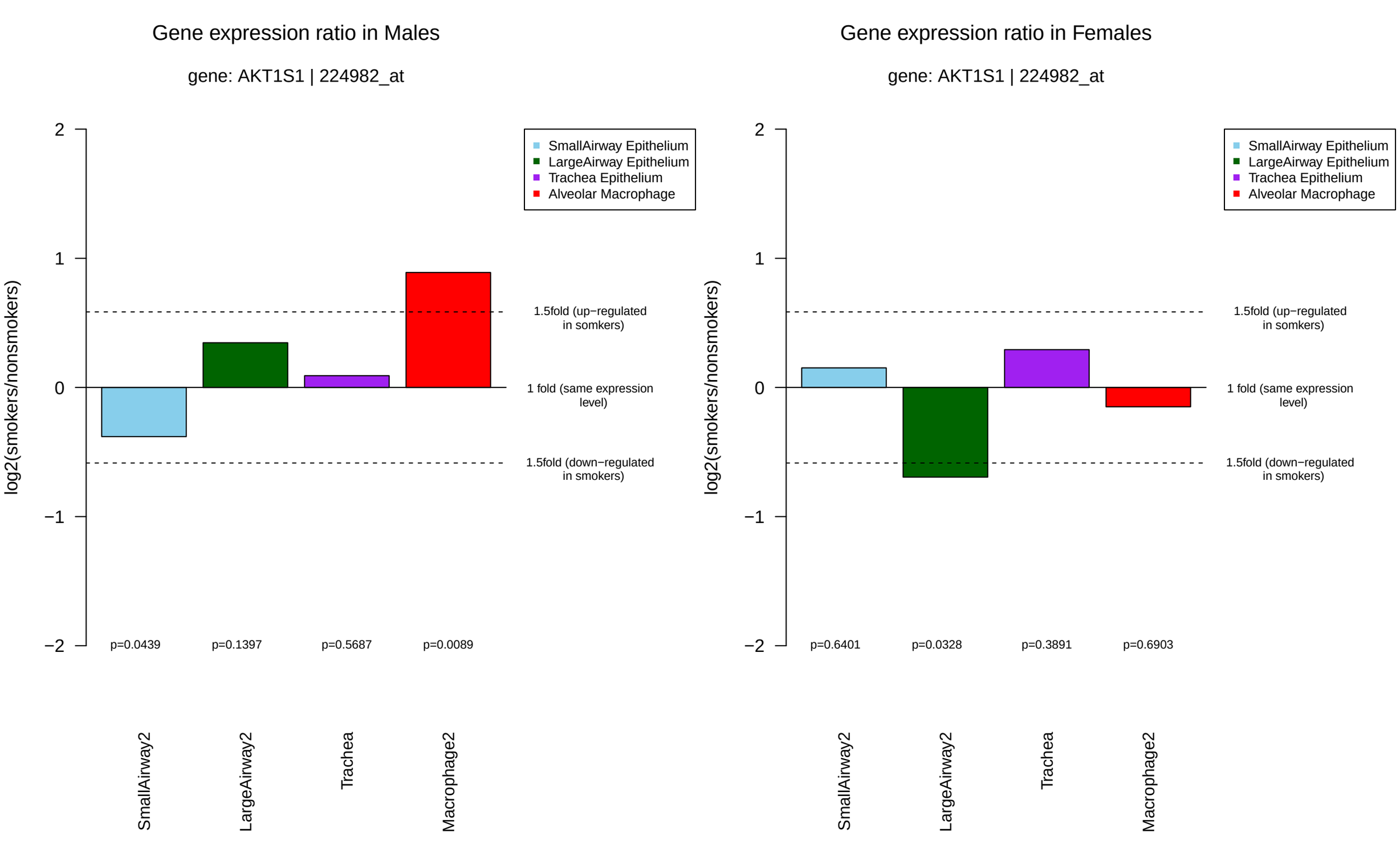
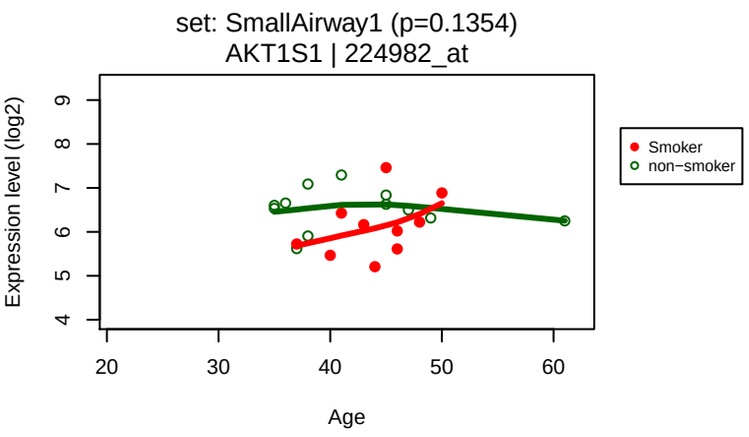
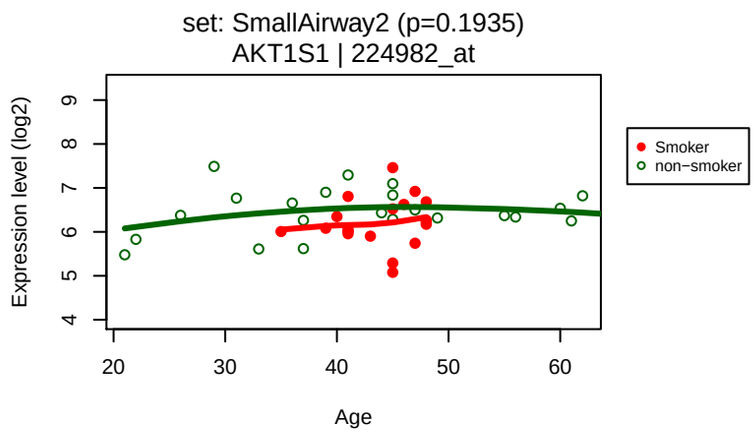
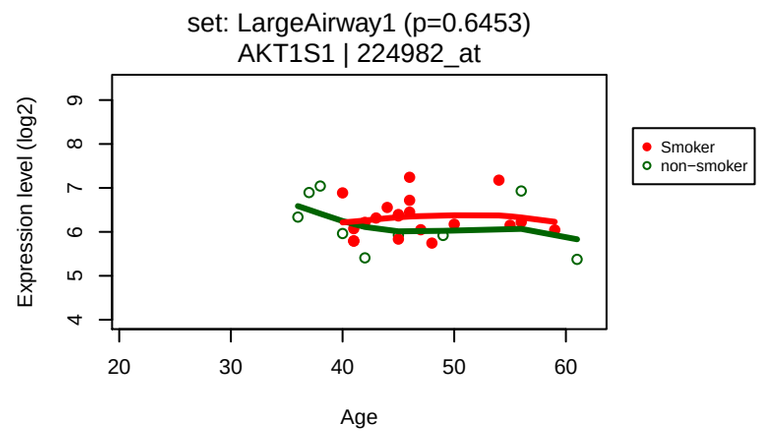

Gene expression of smokers and nonsmokers
gene: AKT1S1 | 224982_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1354
p=0.1935
p=0.4759
p=0.6453
p=0.8721
p=0.4692
p=0.5663
p=0.0317
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

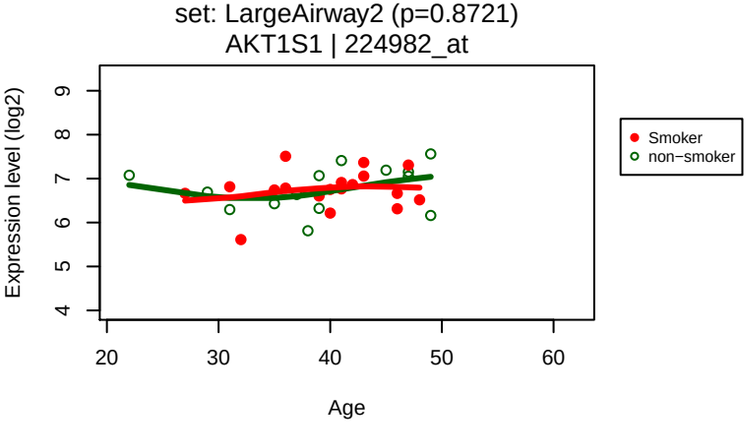

Gene expression ratio (smokers/nonsmokers)
gene: AKT1S1 | 224982_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1354
p=0.1935
p=0.4759
p=0.6453
p=0.8721
p=0.4692
p=0.5663
p=0.0317
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

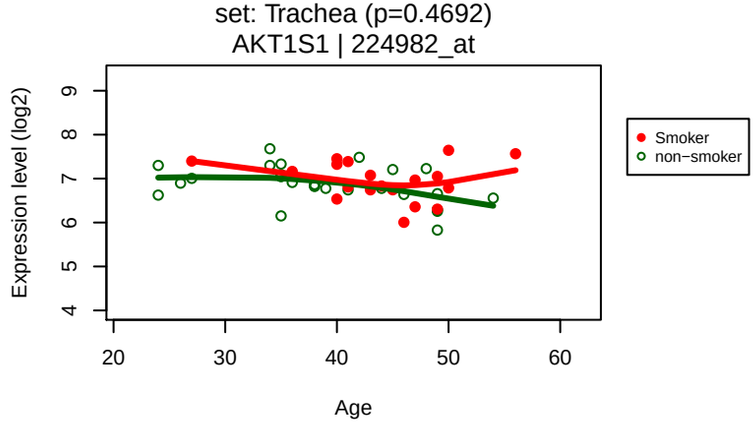

Gene expression ratio in Males
gene: AKT1S1 | 224982_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0439
p=0.1397
p=0.5687
p=0.0089
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKT1S1 | 224982_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6401
p=0.0328
p=0.3891
p=0.6903
SmallAirway2
LargeAirway2
Trachea
Macrophage2

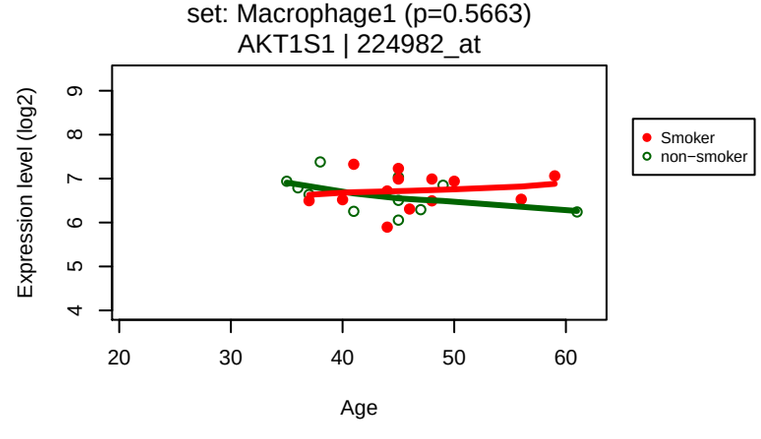

set: SmallAirway1 (p=0.1354)
AKT1S1 | 224982_at
Expression level (log2)
Age
Smoker
non−smoker

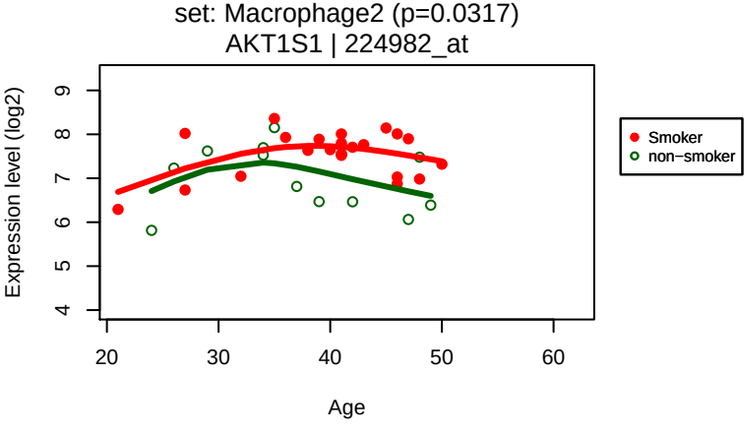

set: SmallAirway2 (p=0.1935)
AKT1S1 | 224982_at
Expression level (log2)
Age
Smoker
non−smoker

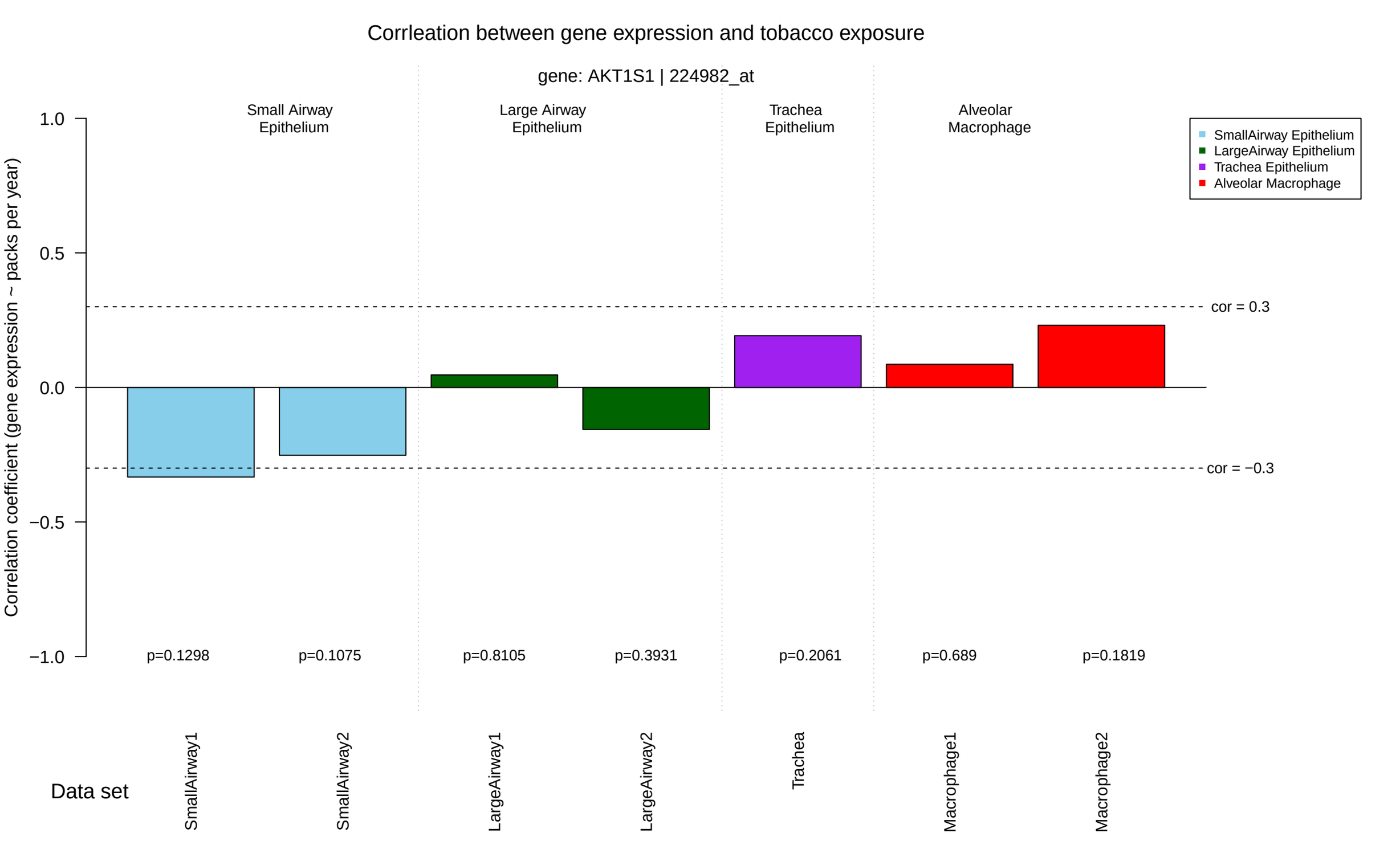

set: LargeAirway1 (p=0.6453)
AKT1S1 | 224982_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8721)
AKT1S1 | 224982_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4692)
AKT1S1 | 224982_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5663)
AKT1S1 | 224982_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0317)
AKT1S1 | 224982_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKT1S1 | 224982_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1298
p=0.1075
p=0.8105
p=0.3931
p=0.2061
p=0.689
p=0.1819
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2